Constancy RoPE: An Empirical Study of Spectral Attention Gains at 4K

Working research report

October 17, 2026

Abstract

We test whether spectral feedback on rotary-position attention gains improves training-free 4K generation with FLUX.1-Krea-dev. Constancy RoPE compares the predicted clean latent spectrum with a native-resolution reference and applies a bounded, noise-gated correction to spatial attention gains. Two completed campaigns fail to demonstrate an improvement over the released SEGA baseline. In the second campaign, only SEGA plus time-matched constancy at gain $\eta = 1$ passes the eight-prompt development gate. On 187 held-out prompt pairs, its ImageReward difference is $+0.00857$ with a 95% confidence interval of $[-0.00841, 0.02742]$, while full-resolution MUSIQ changes by -0.20860 $[-0.70887, 0.31512]$. PickScore and crop MUSIQ decrease. These results do not support an efficacy or state-of-the-art claim. This working report records the completed experiments; it is not a submission-ready methods paper.

1 Question and tested controllers

The project asks whether feedback toward a model's native latent spectrum can improve resolution extrapolation through per-band attention gains. The target image size is 4096×4096, compared with a 1024×1024 native calibration size. The baseline is the released SEGA allocation and RoPE base, identified in the implementation as `sega_official`. The released allocation differs from the paper-equation implementation; the two are not interchangeable baselines.

The first campaign tested the original constancy controller, then a time-matched native spectral target. The latter was also tested on SEGA's RoPE base with its reference temperature. This did not establish an improvement. The second campaign tested two effective-pool temperature rules, `alias_temp` and `alias_temp_sref`, and a composition of released SEGA with time-matched constancy, `sega_ctm`, at $\eta = 0.5$ and 1.0.

For the composed controller, let m_d be SEGA's released query/key multiplier for rotary pair d. A correction in spectral band b is

$$e_b = \tfrac{1}{2}\big(\log P_{\mathrm{native},b}(\sigma) + a - \log \widehat{P}_b\big), \tag{1}$$

$$\delta_b = \eta w_b \ \mathrm{clip}(e_b, -1, 1), \tag{2}$$

$$g_d = m_d^2 \exp\big(\mathrm{interp}(\delta)_d\big). \tag{3}$$

Here a is the anchor offset, w_b is the existing visibility gate, and $\widehat{P}_b$ is measured from the preceding transformer call's clean estimate. The native target is interpolated at the current scheduler noise level. Query and key are each multiplied by $m_d \exp(\delta_d/2)$. The first step has no correction, the index axis retains unit gain, and no additional global temperature is applied. Spectral powers use orthonormal FFTs and means within bands. The vision analogy motivates this controller; it does not establish its efficacy or novelty.

2 Executed protocol

All reported comparisons use the FLUX.1-Krea-dev transformer with FLUX.1-dev text encoders and VAE, 4096^2 output, 28 sampling steps, distilled guidance 4.5, and scheduler shift $\mu = 1.15$. The selected candidate and baseline

share the executed schedule, seed convention, and SEGA RoPE base. Earlier FLUX.1-dev outputs exhibited a lattice artifact and are excluded from these results.

The 195 Aesthetic-4K evaluation captions are split into development rows 0–7 and held-out rows 8–194. Native calibration uses 16 separate Pick-a-Pic prompts. Development eligibility requires positive mean differences in both ImageReward and full-resolution MUSIQ, with CLIP difference at least -0.3. At most two variants may advance; no settings change after selection.

The second campaign reports exactly 187 matched seed-0 prompt pairs. Generation produced 188 images because row 7 was included to preserve task chunking; that row is excluded from all held-out tables. Confidence intervals use 2,000 paired bootstrap draws with bootstrap seed 0 and the 2.5 and 97.5 percentiles. Intervals are nominal per metric, without multiple-comparison adjustment. An interval crossing zero does not establish equivalence.

3 Results

3.1 Development selection

Table 1 records selection only. Both alias variants lose ImageReward. The $\eta = 0.5$ composition misses the MUSIQ gate by 0.007850. Only $\eta = 1$ advances; development scores are not generalization evidence.

Table 1: Development means on eight prompts, seed 0.

Variant	ImageReward	MUSIQ	CLIP	Advance
Released SEGA	1.423018	42.488060	30.320004	Reference
Alias temperature	1.281098	35.807356	29.366986	No
Alias, reference gain	1.277426	44.972845	30.238460	No
SEGA + constancy, $\eta = 0.5$	1.441334	42.480210	30.200250	No
SEGA + constancy, $\eta = 1$	1.448742	42.505928	30.342451	Yes

3.2 Held-out comparison

Table 2: Held-out seed-0 means and paired differences, candidate minus released SEGA, over 187 prompts. Higher is better except the last three rows.

Metric	SEGA	SEGA + constancy	Difference [95% CI]
ImageReward	1.18994	1.19852	$+0.00857$ $[-0.00841, +0.02742]$
PickScore	23.05255	22.97729	-0.07526 $[-0.10234, -0.04914]$
CLIP	29.13552	29.04160	-0.09392 $[-0.19651, -0.00368]$
MUSIQ, full image	45.22784	45.01924	-0.20860 $[-0.70887, +0.31512]$
MUSIQ, 1024 crops	58.32778	57.19612	-1.13166 $[-1.38754, -0.90219]$
CLIP-IQA, full image	0.60299	0.60593	$+0.00294$ $[-0.00018, +0.00609]$
CLIP-IQA, 1024 crops	0.58597	0.58846	$+0.00250$ $[-0.00070, +0.00563]$
Spectrum log RMSE	1.06838	1.06596	-0.00242 $[-0.00905, +0.00433]$
Spectrum slope error	0.19995	0.19741	-0.00254 $[-0.00678, +0.00192]$
Tiling at 1024	0.06836	0.06844	$+0.00008$ $[-0.00072, +0.00081]$

The selected composition has no demonstrated gain on ImageReward or full-resolution MUSIQ. PickScore, crop MUSIQ, and CLIP have negative nominal confidence intervals. The remaining intervals cross zero. This second campaign therefore does not repair the original controller's failure.

The earlier time-matched controller on SEGA's base also regresses on the same 187 seed-0 prompts. Full-resolution MUSIQ changes by -3.03602, with interval $[-4.28169, -1.81488]$. Crop MUSIQ changes by -2.19266, with interval $[-2.81241, -1.56666]$. Its ImageReward difference is unresolved at $+0.01782$ $[-0.02271, 0.05913]$. A temperature-only control on SEGA's base reduces full-resolution MUSIQ by 2.08550 points, with a paired interval $[-3.24526, -0.96407]$. These comparisons constrain the tested designs; they do not show that all spectral controllers must fail.

4 Limits and continuation

The second-campaign candidate has one held-out seed and one evaluated backbone at one resolution. No second-seed replication of this candidate, new backbone, human preference study, or broader resolution claim is supported. Aggregate FID/KID scores from the raw 188-image evaluation include development row 7 and are excluded from the held-out conclusion. No new throughput comparison is claimed in this report.

The earlier theory and literature sections remain in the source tree for review, but are excluded from this snapshot because they describe superseded settings and unresolved claims. They require an implementation and source audit before incorporation into a submission. The next research step is to formulate a different, testable mechanism and freeze a development-only diagnostic before any new GPU campaign. The already inspected held-out set must not become a tuning set presented as fresh evaluation.

Evidence provenance. The completed campaign is recorded at implementation commit `a1ede1a` on October 16, 2026. Its archived paired table, per-prompt scores, and analysis protocol are in `.agent/notes/main2/`. The current held-out table is generated from those CSVs after checking its means and paired intervals against the per-prompt records. This check reuses saved scorer outputs; it is not a fresh image-scoring reproduction. The previous session records 194 passing CPU tests. No inference or training was rerun for this paper update.